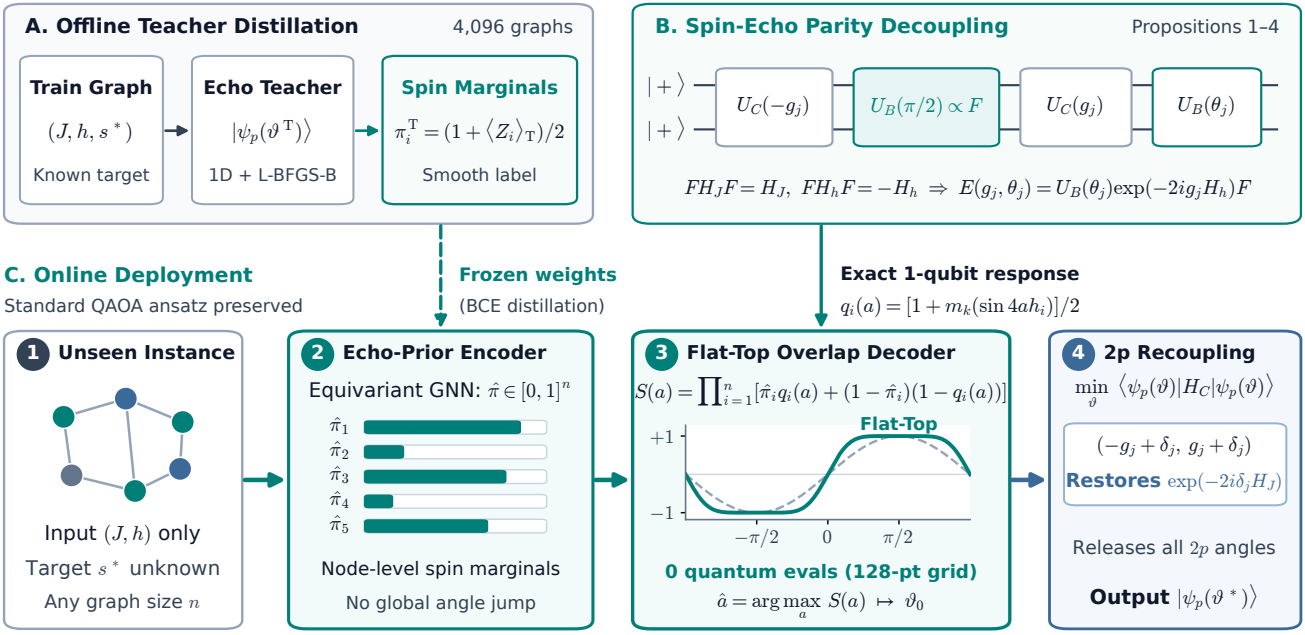

A. Offline Teacher Distillation
4,096 graphs
Train Graph
(J, h, s^*)
Known target
Echo Teacher
$|\psi_p(\vartheta^{\mathrm{T}})\rangle$
1D + L-BFGS-B
Spin Marginals
$\pi_i^{\mathrm{T}} = (1 + \langle Z_i \rangle_{\mathrm{T}})/2$
Smooth label
B. Spin-Echo Parity Decoupling
Propositions 1–4
$|+\rangle$
$|+\rangle$
$U_C(-g_j)$
$U_B(\pi/2) \propto F$
$U_C(g_j)$
$U_B(\theta_j)$
$FH_JF = H_J,\ FH_hF = -H_h \Rightarrow E(g_j, \theta_j) = U_B(\theta_j)\exp(-2ig_jH_h)F$
C. Online Deployment
Standard QAOA ansatz preserved
Frozen weights
(BCE distillation)
Exact 1-qubit response
$q_i(a) = [1 + m_k(\sin 4ah_i)]/2$
1 Unseen Instance
Input (J, h) only
Target s^* unknown
Any graph size n
2 Echo-Prior Encoder
Equivariant GNN: $\hat{\pi} \in [0, 1]^n$
$\hat{\pi}_1$
$\hat{\pi}_2$
$\hat{\pi}_3$
$\hat{\pi}_4$
$\hat{\pi}_5$
Node-level spin marginals
No global angle jump
3 Flat-Top Overlap Decoder
$S(a) = \prod_{i=1}^{n} [\hat{\pi}_i q_i(a) + (1 - \hat{\pi}_i)(1 - q_i(a))]$
Flat-Top
+1
−1
$-\pi/2$
0
$\pi/2$
0 quantum evals (128-pt grid)
$\hat{a} = \arg\max_a S(a) \mapsto \vartheta_0$
4 2p Recoupling
$\min_{\vartheta} \langle \psi_p(\vartheta) | H_C | \psi_p(\vartheta) \rangle$
$(-g_j + \delta_j,\ g_j + \delta_j)$
Restores $\exp(-2i\delta_j H_J)$
Releases all $2p$ angles
Output $|\psi_p(\vartheta^*)\rangle$